

Answers to qualitative questions

General instructions

Your responses should be coherent, clear and precise. Use of bullet points is acceptable.

Task 2.2

Discussion on whether the `contrast0` feature could help distinguish between the `dog\_bark` and `air\_conditioner` classes, with reasoning based on the data (100 words maximum).

Based on the provided violin plot, the contrast0 feature appears to be useful for distinguishing between dog_bark and air_conditioner sounds. The distributions of the two classes exhibit a clear difference in their central tendencies. The dog_bark distribution has a higher median and a greater overall range than the air_conditioner distribution. The dotted lines representing the quartiles for dog_bark are visibly higher, centered around a spectral contrast of 15, whereas those for air_conditioner are lower, centered around 13. This clear separation suggests that the feature could effectively serve as a classifier, despite some overlap.

Task 4.4

Discussion on the following: Compare the accuracy from Task 3 (unscaled) and Task 4 (scaled), where KNN is applied to both unscaled and scaled data, resulting in different performance. In your own words, explain why feature scaling had such an impact on the KNN model's performance (100 words maximum).

Comparing the results, the unscaled KNN model achieved an accuracy of 0.83, while the scaled model's accuracy was 0.88. Although the initial performance was relatively high, feature scaling provided a meaningful improvement. The KNN algorithm's performance is directly tied to its use of distance metrics. Without scaling, features with larger numerical ranges can disproportionately influence the distance calculation, making the model less sensitive to variations in other, smaller-ranged features. Scaling all features to a uniform range ensures that each one contributes equally to the distance metric, enabling the model to make more precise and balanced classifications.

Task 4.5

Discussion on what KNN uses internally that makes it sensitive to the scale of the features (50 words maximum).

KNN internally relies on distance-based calculations, such as Euclidean distance. When features are not scaled, those with a larger numerical range contribute a greater weight to this calculation. This causes the model to become overly sensitive to these features, effectively overshadowing the influence of other, equally important features.

Task 5.2

Compare the performance of the GaussianNB model with the KNN model using unscaled data. Which model performs better, and why? Provide justification for your answer (50 words maximum).

The KNN model performed significantly better on the unscaled data than the GaussianNB model, with accuracies of 0.83 and 0.42, respectively. GaussianNB has a strong assumption that features are conditionally independent and follow a normal distribution. In contrast, KNN

is a non-parametric model that makes no such assumptions about the data's underlying distribution, making it more effective for this dataset.

Task 6.5

For the decision tree, explain why this particular feature was selected as the root node (100 words maximum).

Based on the decision tree plot, the contrast4 feature was chosen as the root node because it yielded the highest Information Gain among all features. Information Gain is a measure of the reduction in entropy (or randomness) that a split provides. The algorithm evaluates every possible feature split and selects the one that provides the largest decrease in uncertainty. Therefore, the tree chose to split on contrast4 as it was the most effective single feature for initially partitioning the data and beginning the classification process.

Task 7.3

Discuss the line plot of the support vector machine in performance accuracy and explain your observations (150 words maximum).

The line plot for the SVM classifier shows a clear trend of increasing accuracy as the regularization parameter C increases. The accuracy rises from approximately 0.709 at $C=0.01$ to 0.734 at $C=0.5$. This observation indicates that the initial models with smaller C values were likely underfitting the data. As the value of C increases, the model is penalized less for misclassifying training data points, allowing it to fit the complexities of the dataset more closely. This improved fit leads to a higher test accuracy. The performance appears to be leveling off, suggesting that a much larger C value may not yield significant further gains.

Task 8.2

Briefly comment on what the confusion matrix reveals about your model's strengths and weaknesses (150 words maximum).

The confusion matrix reveals the model's strengths and weaknesses in classification. Its strengths are evident in its high accuracy for several classes, shown by the large numbers on the diagonal. For example, the model is highly effective at classifying 'jackhammer' (174), 'air_conditioner' (148), and 'drilling' (144), correctly identifying a majority of the instances for these classes.

However, the matrix also highlights the model's weaknesses in distinguishing between certain classes. The most notable confusions occur between acoustically similar sounds. For instance, 'air_conditioner' is frequently misclassified as 'jackhammer' (12), and 'street_music' is often confused with both 'air_conditioner' (15) and 'children_playing' (17). These misclassifications suggest that the model struggles with the feature overlap present in these particular sound types, indicating an area for potential improvement.

Task 8.4

Discuss in what situations F1-score is a more appropriate metric than accuracy (50 words maximum)?

F1-score is a more appropriate metric than accuracy in scenarios involving significantly imbalanced datasets. Accuracy can be misleading in such cases because a model can achieve a high score simply by correctly predicting the majority class. F1-score balances

precision and recall, ensuring that the model performs well on both the majority and minority classes.